Kaustubh Saoji

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Education

Birla Institute of Technology and Science, Pilani *Goa , India*
Master of Science (Hons.) in Biological Sciences *May 2022 - August 2027*
Bachelor of Engineering in Electrical and Electronics Engineering

- **Cumulative GPA:** 7.44/10

Namo Rims International School and College *Pune, India*
Higher Secondary Certificate Examination *December 2020 - June 2022*

- **Percentage:** 84%

Skills

Biology	PCR, qPCR, RNA Extraction, Horseshoe Crab Embryo Handling, SDS Page, Sequence analysis, modeling biological systems, in-silico experimentation, AutoDock, GAMESS.
Machine Learning	Deep learning (PyTorch, TensorFlow), Spiking Neural Networks (SNNs, CSNNs, NeuCube), GNNs, BLSTMs, classical ML (Naive Bayes, MLE, supervised and unsupervised learning).
Programming	Python (NumPy, Pandas, Matplotlib, Seaborn, scikit-learn), PyTorch, Git/GitHub, Linux, HPC.

Research Experience

Undergraduate Thesis Researcher - Generative AI for Peptide Design *West Lafayette, IN, USA*
Purdue Institute for Drug Discovery, Purdue University *August 2026 – Present*

- Conducting thesis research on the de novo design of peptide candidates for immunotherapy.
- Applying generative AI and biomolecular structure models, including Boltz and BoltzGen, to generate and evaluate peptide designs and predicted peptide–target interactions.
- Developing an in silico workflow for peptide generation, structure prediction, and candidate prioritization for downstream validation.

Summer Research Intern *Pilani, India*
CSIR - Central Electronics Engineering Research Institute Pilani *May 2024 – July 2024*

- **Authored** a review paper titled *"Harnessing Cold Atmospheric Plasma for Skin Cancer: A Non-Invasive Breakthrough in Plasma Medicine"*, published in **Drug Discovery Today (Q1 Journal)**. https://doi.org/10.1016/j.drudis.2025.104491
- Investigated apoptotic and molecular pathways in skin cancer, benchmarking conventional therapies against cold plasma–induced apoptosis for improved patient recovery.
- Contributed to the design and development of portable cold plasma prototype devices for direct dermatological applications.

Ab Initio Modeling for Molecular Discovery *Goa, India*
Under Prof Raviprasad Adhuri, BITS Pilani K K Birla Goa Campus *December 2025 - Present*

- Optimized molecular geometries and electronic properties for the QM40/QM9 dataset using GAMESS and Avogadro.
- Identified structural overlaps between QM datasets and FDA-approved drugs via SMILES and InChI identifiers for drug repurposing.

- Executed molecular docking using AutoDock Vina to evaluate and compare the binding affinities of FDA-approved antivirals and phytochemicals against viral targets.

Energy-Efficient SNNs for Seizure Detection *Goa, India*
Under Prof Debasis Das, BITS Pilani K K Birla Goa Campus *May 2025 - Present*

- Applied spike-timing-dependent plasticity (STDP) and surrogate gradient methods for training SNNs on EEG datasets.
- Designed temporal coding schemes to capture spatio-temporal features in EEG signals for robust seizure prediction.
- Evaluated the event-driven processing efficiency of SNNs compared to conventional deep learning models.

High-Purity Transcriptomic Isolation from Horseshoe Crabs *Goa, India*
Under Prof Sumit Biswas, BITS Pilani K K Birla Goa Campus *January 2024 - May 2024*

- The variation of the transcriptome was studied in various stages of horseshoe crab development.
- Achieved a high RNA yield of **300 ng/μL** from *Tachypleus gigas* using a modified CTAB method, outperforming commercial kits and allowing reliable downstream applications.
- Attained an $\mathbf{A_{260}/A_{280}}$ ratio of **2.0**, indicating high purity of RNA and effective protein-free extraction using the modified CTAB method.
- Increased total RNA yield by **500%** using an optimized CTAB method, outperforming the standard Trizol protocol in efficiency.

Awards and Achievements

- Achieved 7th rank globally in the International Rover Challenge 2025, a prestigious Mars rover competition, as the **Science Lead** of BITS Goa's Mars rover team - Project Kratos.
- Awarded the merit cum need scholarship of **40%** on the tuition fees for being in the top **6%** of the batch.
- Ranked in the top **3% nationally** in the Joint Entrance Examination (JEE) Mains, one of the most competitive engineering entrance exams in India.

Position of Responsibility

Science Lead *Goa, India*
Project Kratos - BITS Goa's Mars Rover Team *April 2024 - April 2025*

Led and mentored a team of 50+ members in conducting astrobiology research and developing chemical tests for life detection experiments on the rover.

Designed and implemented machine learning pipelines for autonomous rock detection and classification to aid in planetary exploration tasks.

Student Faculty Member *Goa, India*
Department of Biological Sciences, BITS Pilani K K Birla Goa Campus *October 2024 - May 2025*

Collaborated with peers and senior faculty to represent student concerns and contribute to departmental decision-making processes.

Volunteer Experience

Voluteer Teacher *Goa, India*
Nirmaan, BITS Pilani, K K Birla Goa Campus *December 2022 - May 2024*

Taught middle school mathematics and science to underprivileged children from an under-resourced community near campus, fostering educational access and mentorship.

Mentor *Goa, India*
Academic Assistance Program, BITS Pilani, K K Birla Goa Campus *August 2023 - December 2023*

Mentored first-year students at BITS Goa in the General Biology course by clarifying concepts, addressing doubts, and supporting academic development.

Relevant Courses

Biological Sciences: General Biology, Biochemistry, Microbiology, Cell Biology, Integrated Biology, Molecular Biology, Enzymology, Genetics, Instrumental Methods of Analysis (Lab Course), Bioinformatics, Ecology, Animal Physiology, Plant Physiology, Recombinant DNA Technology, Developmental Biology, Immunology, Biophysics.

Electrical and Electronics Engineering: Electrical Sciences, Computer Programming, Electronic Devices, Digital Design, Electrical Machines, Electromagnetic Theory, Microprocessors and Interfacing, Signals and Systems, Optimisation, Analog and Digital VLSI Design, Communications Systems, Machine Learning.

Miscellaneous: Mathematics I (Multivariable and Vector Calculus), Mathematics II (Linear Algebra and Complex Analysis), Mathematics III (Differential Equations), Probability and Statistics, Thermodynamics, General Chemistry.